AquaWatch

AI-Powered Smart Water Infrastructure Monitoring and Decision Support System

Problem Definition Document

Final Year B.E. Project | Department of Computer Science and Engineering

1. Problem Definition

Rapid urbanization has significantly increased the occurrence of water-related civic issues, including pipeline leakages, waterlogging, sewage overflow, drainage blockages, and damaged water supply infrastructure. These issues adversely affect public health, environmental sustainability, traffic movement, and the quality of urban life. Municipal corporations receive numerous complaints regarding such problems through web portals, mobile applications, and helplines. However, existing municipal complaint management systems primarily focus on complaint registration and basic issue classification using image-based deep learning models.

Although these systems reduce manual complaint categorization, they rely predominantly on single-modal visual inputs and lack intelligent mechanisms for comprehensive complaint analysis. They do not effectively utilize the combined information available in images, textual descriptions, GPS location, and temporal data to understand the complete context of a complaint. Consequently, duplicate complaints are processed independently, increasing administrative workload and leading to inefficient resource utilization. Furthermore, existing systems do not estimate complaint severity, automatically prioritize critical issues, or identify geographical hotspots where recurring water-related problems occur. As a result, municipal authorities often rely on manual analysis and decision-making, which increases response time and reduces the overall efficiency of urban water infrastructure management.

Therefore, there is a need for an AI-powered multimodal water infrastructure monitoring and decision support system that can intelligently analyze citizen complaints by integrating visual, textual, spatial, and temporal information. Such a system should automatically classify water-related issues, detect duplicate complaints, estimate severity, identify geographic hotspots, prioritize complaints based on urgency, and provide actionable insights through an intelligent decision-support dashboard. By enabling data-driven decision-making, the proposed system aims to reduce manual effort, optimize resource allocation, improve response time, and enhance the efficiency of municipal water infrastructure management.

2. Problem Significance and Impact

Water-related civic issues such as pipeline leakages, waterlogging, sewage overflow, drainage blockages, and damaged water supply infrastructure have become increasingly common in rapidly growing urban areas. If these issues are not addressed promptly, they can lead to water wastage, environmental pollution, traffic congestion, property damage, public health risks, and increased maintenance costs. Delays in identifying and resolving such problems also reduce citizens' trust in municipal services and negatively affect the overall quality of urban living.

Although many municipalities have adopted digital complaint management systems, existing solutions primarily automate complaint registration and basic issue classification. They lack intelligent capabilities to identify duplicate complaints, estimate the severity of incidents, detect recurring problem locations, and prioritize complaints based on urgency and impact. Consequently, municipal authorities often spend significant time manually analyzing complaints, resulting in delayed response, inefficient resource allocation, and redundant maintenance activities.

The proposed AquaWatch system addresses these challenges by integrating computer vision, natural language processing, and geospatial analytics to provide intelligent decision support. By automatically classifying water-related issues, identifying duplicate complaints, estimating severity, detecting geographic hotspots, and prioritizing complaints, the system enables municipal authorities to respond more efficiently to critical incidents. This improves resource utilization, reduces response time, minimizes manual effort, and supports data-driven urban infrastructure management.

The successful implementation of AquaWatch can contribute to smarter municipal governance, improved public service delivery, and sustainable urban development. It also aligns with the United Nations Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities by promoting efficient management of urban infrastructure and enhancing the quality of life for citizens.